AI 500 – Assignment 2, Part B: Coding Guide

This guide covers the **coding tasks of Part B** only. For each question it explains what the wording means and what you are expected to produce. It does not give answers, and it does not cover the written interpretation parts. Those must come from your own results.

Useful reference: GeeksforGeeks, *Exploratory Data Analysis in Python*

https://www.geeksforgeeks.org/data-analysis/exploratory-data-analysis-in-python/

Libraries you may use: `pandas`, `numpy`, `matplotlib`, `seaborn`, `scikit-learn`

B1. Understanding and Assessing the Dataset

(a) Dataset structure and quality

- Report how many rows (observations) and columns (variables) there are.
- Sort every column into one of four types: numerical, categorical, binary, temporal. Look at what each column actually contains, not only at how it was loaded.
- Check for missing values, and report the count for each column.
- Check for duplicate records, both fully duplicated rows and repeated IDs. Remember that each row is an appointment, not a patient.

(b) Descriptive statistics

- For the numerical variables, report the full set of summary statistics: **count, mean, standard deviation, minimum, quartiles (25%, 50%, 75%) and maximum**.
- Do this only for columns where these numbers make sense. For example, an average ID number tells you nothing.
- For `Age`, you must **also** report the minimum, Q1, median, Q3, maximum and the **IQR** (IQR = Q3 – Q1). The IQR is not part of the standard summary, so you need to calculate it yourself.

(c) Initial observations

- Write at least two observations about the structure or quality of the data. Support each with a number from your output.

B2. Exploring Distributions and Relationships

"Appropriate visualization" means the **chart type matches the variable type**. There is no single correct chart, but a chart that doesn't suit the data is wrong.

What you want to show	What kind of chart
One numerical variable	A chart that shows its shape and spread
One categorical or binary variable	A chart that compares category counts or proportions
A numerical variable against the target	A chart that compares the distribution across the two outcome groups (attended vs missed)
A categorical variable against the target	A chart that compares the no-show rate across the categories
Many variables at once	A matrix-style plot of pairwise relationships

Your analysis must include:

1. **Distribution of `Age`**
2. **Distribution of at least two categorical or binary variables.** Choose ones that are relevant to the problem.
3. **Relationship between `Age` and `No-show`**
4. **Correlation heatmap.** "Appropriate variables" means numerical and 0/1 variables only. Leave out ID columns and raw text. The target must be turned into a number to be included.

For each plot, write why it is appropriate and one observation from it.

B3. Investigating No-Shows Across Subgroups

(a) Patient characteristics

- Calculate the **overall no-show rate** first.
- Then compare the no-show rate across at least four characteristics. `Gender` and `SMS_received` are required. Pick the other two (or more) yourself.
- Use charts to compare the groups, and show how many patients are in each group.

(b) Age groups

- Divide `Age` into **meaningful groups**: groups that make sense in real life, with boundaries you can justify. Do not use arbitrary cuts.
- Compare the no-show rate of each group.

(c) Intersectional analysis

- "Intersectional" means looking at **two characteristics together** (here Gender and Age Group), not each one separately.
- Calculate the no-show rate for every Gender × Age Group combination and visualize it.

(d) Interpretation

- Written answer, based on your own results.

B4. Temporal Analysis and Feature Engineering

Reference for working with dates: https://www.geeksforgeeks.org/pandas/python-working-with-date-and-time-using-pandas/

(a) Temporal features

- Convert `ScheduledDay` and `AppointmentDay` into proper datetime format.
- Create features such as day of the week, month, and scheduled hour. Check which of the two date columns has a meaningful time of day.

(b) Waiting time

- Create `WaitingDays`: the number of days between the scheduled date and the appointment date.
- Check the values for anything that doesn't make sense, and decide how to handle it.
  - [It means a sanity check: after you create a new column, look at its smallest and largest values and its overall spread, and ask whether each value could be true in real life. For `WaitingDays`, think about what a value like that would mean for when the appointment was booked compared with when it happened. Any values that fail the check need a decision: fix them, remove them, or keep them, and say why (in no more than one sentence). You can do this with the same summary statistics you used in B1(b), a quick count of the odd values, or a chart of the column.]
- Look at the distribution of `WaitingDays`.
- Group the waiting times into sensible groups and compare the no-show rate across them.

(c) Temporal patterns

- "Investigate" means: calculate the no-show rate for each day of the week and for each month, then show the results in a chart (and a small table) so the pattern is easy to see. Include the number of appointments in each group.

(d) Feature usefulness

- Written answer, based on your own results.

B5. Preparing the Dataset for Modelling

Reference for encoding and feature scaling: https://medium.com/@skalthaf505/understanding-data-preprocessing-encoding-and-feature-scaling-0c516022957e

Do these steps in this order:

1. **Encode the target** as a number. Make sure you know which value means "missed".
2. **Select your features.** Decide which columns to keep and which to leave out, and justify the important exclusions.
3. **Split** the data into training and test sets, **before** any scaling or encoding.
4. **One-hot encode** the categorical (nominal) variables.
5. **Min–Max scale** the numerical variables only. Do not scale 0/1 flags or columns that are already encoded.
6. **Fit on training data only.** Every transformation must learn from the training set alone, then be applied to the test set.
7. **Show the result:** the shape and a sample of the final training and test datasets.

Finally, explain in your own words why fitting transformations on the full dataset before splitting causes data leakage.